GUJARAT TECHNOLOGICAL UNIVERSITY

School of Engineering

Official Hackathon Event · 2026

CODESTORM

HACKATHON 2K26

Problem Statement Booklet · For Participant Reference

DURATION

4.5 Hours

3.5 Hrs Coding Window · 1 Hr Evaluation Round

HACKATHON GROUND RULES

- All AI and development tools are permitted during the event
- Direct cloning of public repositories is strictly prohibited
- Original code and authentic implementation are required
- A live working demo is mandatory during evaluation

PROBLEM STATEMENTS OVERVIEW

Four problem statements · Select one to pursue during the event

PS 01

CATEGORY
CYBERSECURITY · ARTIFICIAL INTELLIGENCE

AI-Based Cybersecurity Awareness Solution

Detect phishing and malware threats, educate users in real time, and track awareness scores through an AI-driven classification system.

PS 02

CATEGORY
MOBILE APPLICATION · GENERATIVE AI

Knowledge-Based File Upload & Summarizer (Android)

Build an Android app that accepts documents, generates structured summaries, and supports grounded natural-language Q&A using retrieval-augmented generation.

PS 03

CATEGORY
AI / ML · VIDEO ANALYTICS

Video Transcript, Correlation & Metadata Analysis

Accept video uploads, extract transcript, metadata, and behavioural signals, and surface a real-world insight through correlated timeline analysis.

PS 04

CATEGORY
NATURAL LANGUAGE PROCESSING · COMPUTER VISION

AI-Based OCR Solution for Multi-Foreign Languages

Extract, identify, and translate text from images containing multiple foreign scripts while preserving document layout and structure.

PS 01

CATEGORY

CYBERSECURITY · ARTIFICIAL INTELLIGENCE

SUGGESTED TIME

Approx. 2.0 Hrs

AI-Based Cybersecurity Awareness Solution

Detect threats. Educate users. Strengthen defences — powered by AI.

PROBLEM DESCRIPTION

Build an intelligent, AI-driven platform that educates users about cybersecurity threats in real time. The system must analyse user-submitted inputs — such as suspicious emails, URLs, or messages — and classify them as phishing, malware, social engineering, or safe. It must generate a clear, plain-English explanation of why the content is dangerous, provide actionable advice, and track the user's awareness score across sessions. The solution must be interactive, adaptive, and visually intuitive.

FEATURES TO BUILD

- AI classifier for phishing, malware, and social engineering detection
- Real-time analysis of submitted URLs, emails, and message content
- Plain-English threat explanation with severity rating (Low / Medium / High)
- Actionable recommendation panel for each detected threat type
- User awareness score tracker across sessions with progress graph
- Simulated phishing scenario module for interactive training
- Threat history dashboard with category breakdown and trend chart
- Exportable awareness report (PDF) with user score and findings

AI-Based Cybersecurity Awareness Solution

EXPECTED OUTCOMES

Final submissions should demonstrate each of the following working scenarios

1. Submit a phishing email — the system flags it, explains the indicators, and scores severity
2. Submit a safe URL — the system correctly clears it with reasoning displayed
3. Awareness score increments correctly after each user interaction
4. Simulated phishing scenario runs end to end, guiding the user through the red flags
5. Exported PDF contains a readable threat summary and user score breakdown

PS 02

CATEGORY

MOBILE APPLICATION · GENERATIVE AI

SUGGESTED TIME

Approx. 2.5 Hrs

Knowledge-Based File Upload & Summarizer

Upload any document. Build a knowledge base. Query it instantly — on Android.

PROBLEM DESCRIPTION

Build an Android application that allows users to upload knowledge files — PDFs, DOCX, or TXT — and automatically generates a structured summary containing a TL;DR, key concepts, important terms, and actionable insights. Users can then query the document in natural language through an in-app chat interface. All answers must be grounded in the uploaded content — the system must not hallucinate. The app must function entirely offline after initial model load, or clearly declare its API dependency.

FEATURES TO BUILD

- Android app with file picker supporting PDF, DOCX, and TXT formats
- Automatic structured summary — TL;DR, key concepts, and terminology glossary
- In-app conversational Q&A grounded in the uploaded document (RAG-based)
- Out-of-scope question detection — responds 'Not covered in this document'
- Knowledge base management — upload multiple files and switch between them
- Keyword and concept highlighting on the source document view
- Export summary as a shareable PDF from within the app
- Offline-capable processing after initial setup, or clearly declared API mode

Knowledge-Based File Upload & Summarizer

EXPECTED OUTCOMES

Final submissions should demonstrate each of the following working scenarios

1. Upload a 10-page PDF — structured summary is generated within 30 seconds
2. Ask 'What are the key terms?' — app returns an accurate glossary from the document
3. Ask an out-of-scope question — app responds correctly with 'Not covered'
4. Switch between two uploaded documents — Q&A context switches correctly
5. Export summary PDF — output is readable and formatted on the device

PS 03

CATEGORY

AI / ML · VIDEO ANALYTICS

SUGGESTED TIME

Approx. 3.0 Hrs

Video Transcript, Correlation & Metadata Analysis

Upload a video. Extract every signal. Deliver a real-world insight from within.

PROBLEM DESCRIPTION

Build a platform that accepts video uploads and performs multi-layer analysis: automatic transcription, extraction of video metadata (duration, resolution, frame rate, timestamps), behavioural parameters (speaker pace, pause frequency, sentiment arc), and semantic correlation between spoken content and visual scene changes. The system must identify key events in the video, produce a structured timeline, and surface a real-world insight — such as flagging lecture engagement drops, detecting policy violations in recorded meetings, or identifying product demonstration moments in sales calls.

FEATURES TO BUILD

- Video upload supporting MP4, MOV, AVI with automatic transcription
- Metadata extraction — duration, resolution, FPS, bitrate, chapter markers
- Behavioural parameter analysis — speaking pace, pause frequency, sentiment arc
- Scene-change detection correlated with transcript segments and timestamps
- Structured timeline — key events, topic shifts, and emotional peaks mapped to the video
- Real-world insight output — engagement drop detection, violation flags, or key-moment tagging
- Interactive video player with clickable timeline markers and transcript sync
- Downloadable JSON and PDF report with all extracted parameters and insights

Video Transcript, Correlation & Metadata Analysis

EXPECTED OUTCOMES

Final submissions should demonstrate each of the following working scenarios

1. Upload a 10-minute video — full transcript with timestamps is generated correctly
2. Scene changes are detected and aligned with the corresponding transcript segments
3. Behavioural parameters — speaking pace and sentiment arc — are plotted across the timeline
4. A real-world insight is surfaced (for example, an engagement drop flagged with reasoning)
5. Exported report contains the timeline, parameters, and insight in a readable format

PS 04

CATEGORY

NATURAL LANGUAGE PROCESSING · COMPUTER VISION

SUGGESTED TIME

Approx. 3.0 Hrs

AI-Based OCR Solution for Multi-Foreign Languages

Point at any text in any language. Extract, translate, and structure it instantly.

PROBLEM DESCRIPTION

Build an AI-powered OCR system that accepts images or scanned documents containing text in multiple foreign languages — including non-Latin scripts such as Arabic, Chinese, Japanese, Korean, Devanagari, and Cyrillic — and accurately extracts the text, identifies the source language, translates it to English, and structures the output into a clean, readable format. The system must handle mixed-language documents (multiple languages on the same page), preserve document layout structure where possible, and provide a confidence score for each extracted segment.

FEATURES TO BUILD

- Image or PDF upload with OCR for Latin and non-Latin scripts (Arabic, CJK, Devanagari, Cyrillic)
- Automatic language detection per text block with confidence score displayed
- Translation to English, with optional back-translation to a user-selected language
- Mixed-language document support — handles multiple scripts on the same page
- Layout preservation — paragraph structure, headings, and table cells maintained
- Side-by-side view of original image with bounding-box highlights and extracted text
- Structured output — JSON with language, confidence, original text, and translation per block
- Export full document as a translated PDF with original layout preserved

AI-Based OCR Solution for Multi-Foreign Languages

EXPECTED OUTCOMES

Final submissions should demonstrate each of the following working scenarios

1. Upload an Arabic document — text is extracted, language identified, and translated to English
2. Upload a mixed-language image — both blocks are extracted and labelled separately
3. Confidence scores are displayed per block, averaging at least 80% on clear text
4. Structured JSON output is downloaded with all blocks, languages, and translations
5. Exported translated PDF preserves the original layout and remains readable